

5(a)	statement	discontinuous variation	continuous variation	4
	often involves one gene only	✓		
	environmental factors may affect gene expression		✓	
	there is an additive effect of genes that contributes to the phenotype		✓	
	there are distinct differences between the various forms of a characteristic	✓		
5(b)(i)	ref. to spread of results about the <u>mean</u> ;			1

Question	Answer	Marks
5(b)(ii)	$t = \underline{4.255} ;;;$ give 2 marks: if answer is 4.25 or 4.26 if answer is more than 4 significant figures e.g. 4.255439871 if answer is incorrectly rounded to 4 significant figures e.g. 4.256 if answer incorrect then max 2 for working numerator = 4.2 denominator = 0.986972 ecf - if no working mark given then allow one mark for 12.02 as the answer	3
5(b)(iii)	1 there is a higher (mean) concentration of serum 25-OHD in summer ; ora any two from: 2 value of $t / 4.255 > \text{critical value} / 3.773$; 3 there is a <u>significant difference</u> between the two means ; 4 any <u>difference</u> is not due to chance / <u>difference</u> due to chance less than 0.0001% ; ecf from incorrect answer to 5bii	3
5(b)(iv)	genetic AND environmental OR an example of a genetic cause AND an example of an environmental cause ;	1